
Chapter 12: Factorisation

Worksheet 12: Factorising Algebraic Expressions

Learning Objective: Students will factorise algebraic expressions by taking common factors, by grouping, using identities ($a^2 - b^2$, perfect squares), and factorise simple quadratic trinomials; divide a polynomial by a monomial or binomial.

Worked Example

Factorise: $6x^2y + 9xy^2 - 12xy$

Solution: HCF = $3xy$
 $= 3xy(2x + 3y - 4)$

Factorise: $x^2 + 7x + 12$

Find factors of 12 that add to 7: $3 \times 4 = 12$, $3 + 4 = 7$
 $= (x + 3)(x + 4)$

Questions 1–5: Easy / Recall

Q1. Factorise: $4x + 8y$

Q2. Factorise: $x^2 - 25$

Q3. Factorise: $a^2 + 6a + 9$

Q4. Factorise: $3pq + 6p^2q$

Q5. Divide: $(12x^2 - 8x) \div 4x$

Questions 6–10: Basic Application

Q6. Factorise: $x^2 + 5x + 6$

Q7. Factorise: $y^2 - y - 12$

Q8. Factorise by grouping: $ax + ay + bx + by$

Q9. Factorise: $4m^2 - 36n^2$

Q10. Divide: $(x^2 + 7x + 12) \div (x + 3)$

Questions 11–15: Practice and Skill Building

Q11. Factorise: $2a^2 - 50$

Q12. Factorise: $x^2 - 4x - 21$

Q13. Factorise by grouping: $2xy + 3y + 2xz + 3z$

Q14. Divide: $(4x^3 - 8x^2 + 12x) \div 4x$

Q15. Factorise completely: $x^4 - 81$

Questions 16–18: Higher Order Thinking

Q16. Find the value of k so that $(x^2 + 5x + k)$ has $(x + 3)$ as a factor. Then find the other factor.

Q17. Factorise: $a^2(b - c) + b^2(c - a) + c^2(a - b)$. (Hint: factorise by grouping.)

Q18. Using factorisation, evaluate $997^2 - 3^2$ without a calculator.

Questions 19–20: Real-Life Applications

Q19. The area of a rectangular field is $(x^2 + 11x + 28)$ sq. m. Factorise to find the possible dimensions of the field. If $x = 5$, find the actual area, length, and breadth.

- Q20.** A tiler needs to cover a floor of area $(6x^2 + 17x + 12)$ sq. m using rectangular tiles. Factorise the expression to find possible tile dimensions. If $x = 2$, find actual tile size.

Reflection Question: Factorisation is the reverse of expansion. Explain with an example. Why is factorisation useful in simplifying algebraic fractions?